

object
state

CSCI
134

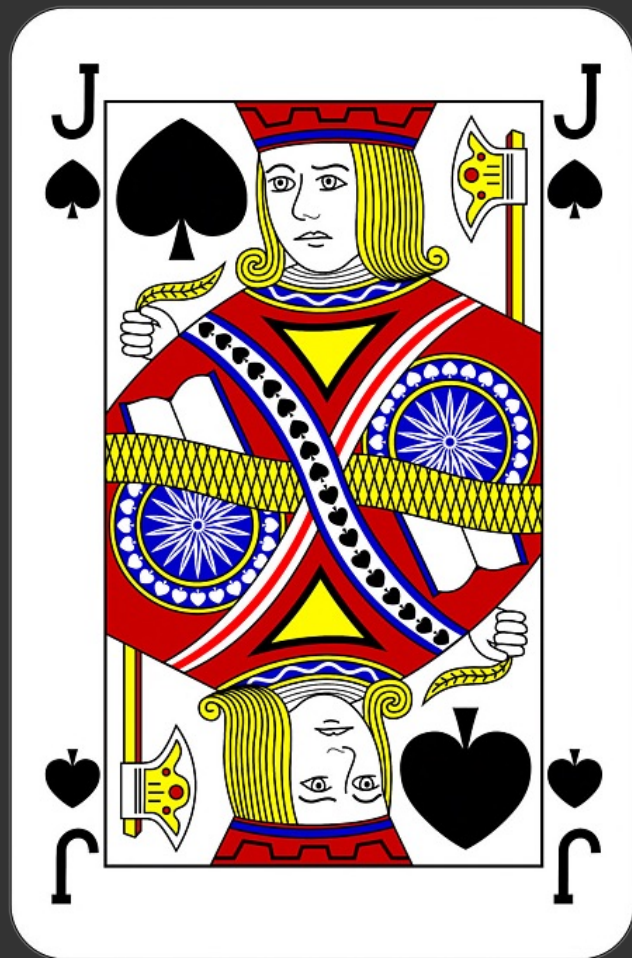


remain in
~~let's go to~~
vegas

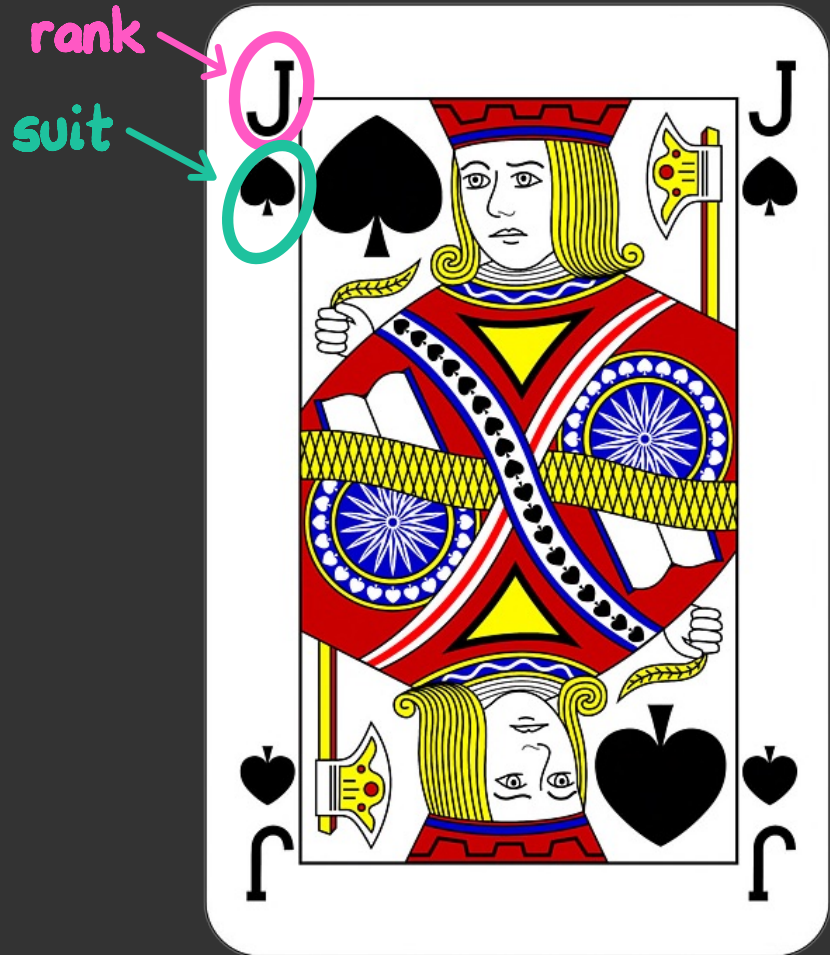
but move to the
blackjack table



how might we
create a type for
playing cards?



cards have two key
attributes:
rank and suit



we'd like to be
able to do this

```
In [1]: from vegas import Card
```

```
In [2]: card1 = Card()
```

```
In [3]: card1.initialize('K', 'clubs')
```

```
In [4]: card1.to_str()
```

```
Out[4]: 'K♣'
```

```
In [5]: card2 = Card()
```

```
In [6]: card2.initialize('5', 'hearts')
```

```
In [7]: card2.to_str()
```

```
Out[7]: '5♥'
```

but we need a
way to associate
values
with objects

```
In [1]: from vegas import Card
In [2]: card1 = Card()
In [3]: card1.initialize('K', 'clubs')
In [4]: card1.to_str()
Out[4]: 'K♣'
In [5]: card2 = Card()
In [6]: card2.initialize('5', 'hearts')
In [7]: card2.to_str()
Out[7]: '5♥'
```

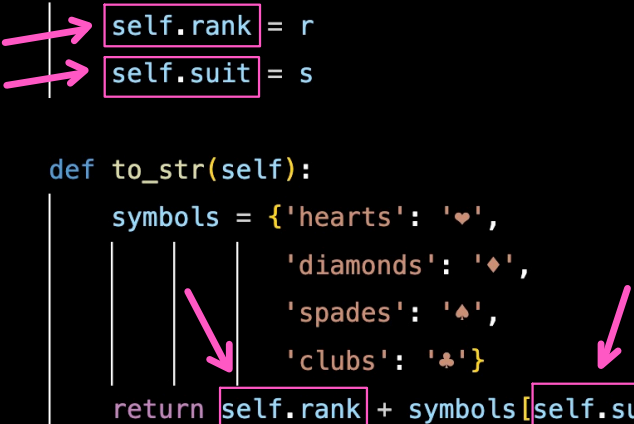
has to know its
rank and suit

The diagram consists of two curved arrows. The first arrow starts from the 'clubs' argument in the 'initialize' call of In [3] and points to the 'card1' object in the 'to_str' call of In [4]. The second arrow starts from the 'hearts' argument in the 'initialize' call of In [6] and points to the 'card2' object in the 'to_str' call of In [7].

we can associate a value with an object
using **attributes** (a.k.a. **instance variables**)

```
class Card:
    def initialize(self, r, s):
        self.rank = r
        self.suit = s

    def to_str(self):
        symbols = {'hearts': '♥',
                  'diamonds': '♦',
                  'spades': '♠',
                  'clubs': '♣'}
        return self.rank + symbols[self.suit]
```



```
In [1]: from vegas import Card
```

```
In [2]: card1 = Card()
```

```
In [3]: card1.initialize('K', 'clubs')
```

```
In [4]: card1.to_str()
```

```
Out[4]: 'K♣'
```

```
In [5]: card2 = Card()
```

```
In [6]: card2.initialize('5', 'hearts')
```

```
In [7]: card2.to_str()
```

```
Out[7]: '5♥'
```


attributes are
just variables
associated with
objects

```
In [1]: from vegas import Card
```

```
In [2]: card1 = Card()
```

```
In [3]: card2 = Card()
```

```
In [4]: card1.suit = "clubs"
```

```
In [5]: card2.suit = "hearts"
```

```
In [6]: card1.suit
```

```
Out[6]: 'clubs'
```

```
In [7]: card2.suit
```

```
Out[7]: 'hearts'
```